

# Panrui Ni, PhD

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Birth: July 16, 1996



## Research Interests

📖 Hamiltonian Dynamics, Hamilton-Jacobi Equations, Aubry-Mather Theory, Weak KAM Theory.

## Academic Position

2024 – present      **Visiting researcher, Graduate School of Mathematical Sciences, The University of Tokyo, Tokyo, Japan.**

Research Project: *Homogenization theory for Hamilton-Jacobi equations.*

Mentor: Hiroyoshi Mitake

2023 – 2024      **Postdoc Faculty, Sorbonne Université, CNRS, IMJ-PRG, Paris, France.**

Research Project: *Discrete and continuous weak KAM theory.*

Mentor: Maxime Zavidovique

## Education

2018 – 2023      **PhD, Fudan University, Shanghai, China** Mathematics.

Thesis title: *Viscosity solutions of contact-type Hamilton-Jacobi equations.*

Advisor: Jun Yan

2014 – 2018      **Bachelor, Southeast University, Nanjing, China** Engineering Mechanics.

Thesis title: *Variational principle for contact Hamiltonian systems and its applications.*

Advisor: Changwen Mi

## Research Publications

### Journal Articles

1      **Panrui Ni**, “Multiple asymptotic behaviors of solutions in the generalized vanishing discount problem,” *Proceedings of the American Mathematical Society*, vol. 151, pp. 5239–5250, 2023, 🔗 URL: <https://doi.org/10.1090/proc/16420>.

2      **Panrui Ni**, “Time periodic solutions of first order mean field games from the perspective of Mather theory,” *Journal of Differential Equations*, vol. 412, pp. 881–901, 2024, 🔗 URL: <https://doi.org/10.1016/j.jde.2024.09.006>.

- 3 **Panrui Ni**, “Weakly coupled Hamilton-Jacobi systems without monotonicity condition: A first step,” *Communications on Pure and Applied Analysis*, vol. 23, no. 7, pp. 961–983, 2024,  URL: <https://doi.org/10.3934/cpaa.2024042>.
- 4 **Panrui Ni**, K. Wang, and J. Yan, “A weakly coupled mean field games model of first order for  $k$  groups of major players,” *Proceedings of the American Mathematical Society*, published online,  URL: <https://doi.org/10.1090/proc/16342>.
- 5 **Panrui Ni**, K. Wang, and J. Yan, “Viscosity solutions of contact Hamilton-Jacobi equations with Hamiltonians depending periodically on unknown functions,” *Communications on Pure and Applied Analysis*, vol. 22, no. 2, pp. 668–685, 2023,  URL: <http://doi.org/10.3934/cpaa.2023005>.
- 6 **Panrui Ni** and L. Wang, “A nonlinear semigroup approach to Hamilton-Jacobi equations–revisited,” *Journal of Differential Equations*, vol. 403, pp. 272–307, 2024,  URL: <https://doi.org/10.1016/j.jde.2024.05.039>.
- 7 **Panrui Ni** and L. Wang, “Aubry-Mather theory for contact Hamiltonian systems III,” *Science China Mathematics*, published online,  URL: <https://link.springer.com/article/10.1007/s11425-022-2197-4>.
- 8 **Panrui Ni**, L. Wang, and J. Yan, “A representation formula of the viscosity solution of the contact Hamilton-Jacobi equation and its applications,” *Chinese Annals of Mathematics, Series B*, to appear,  URL: <https://arxiv.org/abs/2101.00446>.
- 9 **Panrui Ni** and B. Shen, “On variation of action integral in Finsler gravity,” *Annals of Physics*, vol. 404, no. 1, pp. 93–114, 2019.  URL: <https://doi.org/10.1016/j.aop.2019.02.009>.

## Preprints

- 1 A. Davini, **Panrui Ni**, J. Yan, and M. Zavidovique, “Convergence/divergence phenomena in the vanishing discount limit of Hamilton-Jacobi equations.”  URL: <https://arxiv.org/abs/2411.13780>.
- 2 H. Mitake and **Panrui Ni**, “Quantitative homogenization of convex Hamilton-Jacobi equations with Neumann type boundary conditions.”  URL: <https://arxiv.org/abs/2501.12013>.
- 3 H. Mitake and **Panrui Ni**, “Rate of convergence for homogenization of nonlinear weakly coupled Hamilton-Jacobi systems.”  URL: <https://arxiv.org/abs/2412.06428>.
- 4 **Panrui Ni** and L. Wang, “On Mather’s Lipschitz graph theorem of the Aubry set for contact Hamiltonian systems,” submitted.
- 5 **Panrui Ni** and M. Zavidovique, “Nonlinear and degenerate discounted approximation in discrete weak KAM theory.”  URL: <https://arxiv.org/abs/2403.04563>.

## Skills

Languages      Chinese (Native), English (Fluent).

## Skills (continued)

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Software      Mathematica & Python

## Miscellaneous Experience

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### Scholarships and Grants

- 2023      Award of Outstanding Graduate of Shanghai
- 2022      National Natural Science Foundation of China, **participant**, Grant No. 12171096.
- 2021      Qinghua Scholarship at School of Mathematical Sciences, Fudan University
- 2020      Academic Scholarships for PhD Degree Students
- 2019      National Scholarship & Outstanding Student of Fudan University

### Conference Activities

- 2024.1      ANR meeting, École Normale Supérieure de Lyon, **Invited speaker**.  
Title: *On discrete nonlinear vanishing discount problem*.
- 2023.6      PDE seminar, University of Tokyo, **Invited speaker**.  
Title: *Hamilton-Jacobi equations depending Lipschitz continuously on the unknown function*.
- 2022.7      Conference on Differential Equations and Dynamical Systems, Beijing Institute of Technology, **Invited speaker**.  
Title: *A nonlinear semigroup approach to a class of nonmonotone Hamilton-Jacobi equations*.

### Teaching Activities

- 2021      Teaching assistant in Fudan University, Course: *Calculus*.
- 2020      Teaching assistant in Fudan University, Course: *Classical Mechanics*.
- 2019      Teaching assistant in Fudan University, Course: *Classical Mechanics*.